

Jatin Singh Shekhawat

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[Portfolio](#) [LinkedIn](#) [GitHub](#)

ACADEMIC DETAILS

- BK Birla Institute of Engineering and Technology, Pilani (2022–26) – B Tech ECE (8.43 CGPA)
- Star Academy, Jhunjhunu (2021–22) – Senior Secondary (RBSE)
- Star Academy, Jhunjhunu (2019–20) – Matriculation (RBSE)

FIELDS OF INTEREST

- Embedded Systems, IoT, Hardware-Software Co-Design for automation, wireless control, and smart device communication

TECHNICAL SKILLS

- **Programming language:** C, Python
- **Development Tools IDEs:** Arduino IDE, ModelSim, Xilinx, TCLsh, EDA Playground, VS Code
- **Software for Projects:** Inkscape, Universal Gcode Sender, Drawbot
- **Communication Protocols:** UART, Wi-Fi (ESP8266), IR Remote, Bluetooth (HC-05)
- **Areas of Expertise:** Embedded Systems, IoT Integration, G-code Automation, Real-time Control Systems

CERTIFICATES

- Achieved proficiency in IoT Embedded Systems (Coursera). [view certificate](#)

MAJOR PROJECTS AND SEMINAR

- **Voice-Activated CNC Machine (Q2 2024)** – Developed handwriting CNC machine with Arduino, ESP8266, stepper motors, 3D-printed parts, integrated with Google Assistant & IFTTT for wireless G-code execution.
- **IR-Based Wireless Transmitter (Jun 2025)** – Arduino IR transmitter using IRLib2 for AC/fan remotes, enabled custom serial control for home automation.
- **Reflow Oven Temperature Controller (Q3 2025)** – Arduino PID controller using thermistor, Peltier, OLED display; automated soldering stages.
- **Smart Energy Meter with Load Control (Q3 2025)** – ESP32-based energy meter using sensors, relays, MOSFETs, AMS1117, real-time monitoring & load control.

TRAINING AND INTERNSHIPS

- **Advanced Digital Design using HDL – FutureWiz (Jun–Aug 2024)** [view certificate](#) – 45-day training on HDL-based digital design, synthesis, simulation, verification.
- **Summer Internship – CSIR-CEERI, Pilani (Jun–Jul 2025)** [view certificate](#) – Worked on Arduino-based PID Temperature Controller for Reflow Oven; experience in control systems & thermal automation.